

Kisan Mitra AI - Smart Crop Recommendation Report

Farmer Email:	test@farmer.com	Season:	Kharif
State:	Maharashtra	District:	Pune
Soil Type:	Black Soil	Irrigation:	Medium
Water Source:	Medium	Farm Size:	N/A
Previous Crop:	N/A	Date:	16 June 2026

Seasonal Suitability Analysis

Overall Suitability Score: 88%

Agricultural analysis for Pune, Maharashtra during Kharif (Monsoon/June-October) season. The region's Black Soil is excellent for cultivation. With Medium irrigation availability and Medium water source, the farm is well-positioned for moderate crops.

General Advice: For Black Soil in Pune, ensure proper soil testing before sowing. Maintain balanced irrigation schedules. Maintain soil organic matter with green manuring and compost application.

Critical Alerts

- Monitor local weather advisories for Maharashtra during Kharif season for unexpected rainfall or temperature changes.
- Ensure seed treatment with fungicide before sowing to protect against soil-borne diseases common in Black Soil.

Recommended Crops

Maize (Corn) (Suitability Score: 90% | Water: Medium | Demand: High | Growing Period: 90-110 days)

Why Recommended:	Fast-growing cereal crop with diverse uses (food, feed, industrial starch). Adapts well to various soil types and gives good returns in Kharif season. Well-suited for Prakasam, Andhra Pradesh region with Red Soil. Well-suited for Chittoor, Andhra Pradesh region with Black Soil. Well-suited for Pune, Maharashtra region with Black Soil.
Water Management:	Maize needs moderate water with critical irrigation at tasseling and grain-filling stages. Avoid waterlogging as it damages root systems. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply DAP (50 kg/acre) and MOP (30 kg/acre) as basal. Top dress with Urea (40 kg/acre) at knee-high stage and tasseling.
Expected Yield:	20-30 quintals/acre

Cotton (Suitability Score: 88% | Water: Medium | Demand: High | Growing Period: 150-180 days)

Why Recommended:	Major commercial cash crop with strong market linkages. Performs best in black cotton soils with warm temperatures above 20°C. Well-suited for Prakasam, Andhra Pradesh region with Red Soil. Well-suited for Chittoor, Andhra Pradesh region with Black Soil. Well-suited for Pune, Maharashtra region with Black Soil.
Water Management:	Cotton requires moderate and well-timed irrigation. Critical water needs are during flowering and boll formation. Excess water causes boll rot. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply FYM (5 tonnes/acre), DAP (50 kg/acre) as basal, Urea (30 kg/acre) at squaring stage, and spray micronutrients (Boron, Zinc).
Expected Yield:	8-12 quintals/acre (seed cotton)

Pigeon Pea (Arhar/Tur Dal) (Suitability Score: 85% | Water: Medium | Demand: High | Growing Period: 140-180 days)

Why Recommended:	Essential pulse crop for protein security. Fixes atmospheric nitrogen, improving soil fertility for subsequent crops. High MSP support from government. Well-suited for Prakasam, Andhra Pradesh region with Red Soil. Well-suited for Chittoor, Andhra Pradesh region with Black Soil. Well-suited for Pune, Maharashtra region with Black Soil.
Water Management:	Pigeon pea is drought-tolerant but responds well to 2-3 irrigations during flowering and pod development if rainfall is insufficient. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Apply Rhizobium seed treatment, DAP (40 kg/acre) as basal. Minimal nitrogen needed due to biological fixation. Spray 2% Urea foliar at flowering.
Expected Yield:	6-10 quintals/acre

Soybean (Suitability Score: 85% | Water: Low | Demand: High | Growing Period: 90-110 days)

Why Recommended:	Called 'Golden Bean' for its high protein and oil content. Excellent for rain-fed farming with short duration. Fixes nitrogen and improves soil health. Well-suited for Prakasam, Andhra Pradesh region with Red Soil. Well-suited for Chittoor, Andhra Pradesh region with Black Soil. Well-suited for Pune, Maharashtra region with Black Soil.
Water Management:	Soybean is well-suited for rain-fed conditions and can manage with minimal supplemental irrigation. Ensure good drainage as it is sensitive to waterlogging. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability. Your farm has Medium irrigation and Medium water source availability.
Fertilizer Plan:	Rhizobium + PSB seed inoculation. Apply DAP (40 kg/acre) as basal. Avoid excessive nitrogen as it inhibits nodulation.
Expected Yield:	8-12 quintals/acre

■ High Risk / Unsuitable Crops

Wheat: Wheat is a cool-season Rabi crop requiring temperatures of 15-25°C. Kharif season's high heat (30-40°C) and humidity cause poor germination, excessive vegetative growth, and severe rust/blight diseases.

Mustard: Mustard requires cool, dry winters for proper seed formation. Planting in Kharif causes excessive vegetative growth, flower drop due to heat, and white rust/Alternaria blight in humid conditions.